

Adversarial Evaluation of Certificate Revocation Schemes in V2X Communications

A Simulation-Based Robustness Comparison of Active, Passive, and Self-Revocation Under Attack

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Master thesis submitted under the supervision of
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in order to be awarded the Degree of
Master in Cybersecurity
System Design

Academic year
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Abstract

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Preface

"I solemnly swear that I'm up to good" (incorrectly inspired by J.K. Rowling)

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List of Abbreviations

AA	Authorization Authority
AT	Authorization Ticket
AV	Autonomous Vehicle
BTP	Basic Transport Protocol
C-ITS	Cooperative Intelligent Transport Systems
C-V2X	Cellular Vehicle-to-Everything
CAM	Cooperative Awareness Message
CRL	Certificate Revocation List
DAA	Direct Anonymous Attestation
DENM	Decentralized Environmental Notification Message
DSRC	Dedicated Short-Range Communications
EA	Enrolment Authority
ETSI	European Telecommunications Standards Institute
GNSS	Global Navigation Satellite System
HB	Heartbeat
IEEE	Institute of Electrical and Electronics Engineers
ITS	Intelligent Transport System
ITS-G5	ITS communications standard for the 5.9 GHz band
ITS-S	ITS Station
LTE	Long Term Evolution
MAC	Medium Access Control
MDS	Misbehaviour Detection System
OBU	On-Board Unit
OMNeT++	Objective Modular Network Testbed in C++
OSR	Order for Self-Revocation
PKI	Public Key Infrastructure
PRL	Pending Revocation List
RA	Revocation Authority
RCA	Root Certificate Authority
REWIRE	Revocation Without Resolution
RF	Radio Frequency
RSU	Roadside Unit
SCMS	Security Credential Management System
SHB	Single-Hop Broadcast
SUMO	Simulation of Urban MObility
TC	Trusted Component
TEE	Trusted Execution Environment
TLS	Transport Layer Security
TPM	Trusted Platform Module
UTC	Coordinated Universal Time
V2X	Vehicle-to-Everything

Chapter 1

Introduction

Intelligent Transport Systems (ITS) are fundamentally reshaping modern mobility by targeting road transportation’s most persistent challenges such as traffic congestion and fatal accidents. At the core of ITS is Vehicle-to-Everything (V2X) communication—the underlying connectivity engine that enables vehicles to exchange real-time operational data with their surrounding environment.

Over recent years, V2X has rapidly transitioned from an experimental research domain into active large scale deployment [4]. Concurrently, Autonomous Vehicles (AVs) are moving beyond closed testbeds into public roadway trials worldwide. Driven by breakthroughs in artificial intelligence, sensor fusion, and cooperative networking, the global autonomous vehicle market is projected to grow exponentially from USD 3.36 Trillion in 2026 to USD 41.75 in 2034 [15].

As automated driving platforms increasingly rely on shared communication to navigate complex environments, the necessity for secure and trustworthy V2X networks becomes paramount. In a cooperative driving ecosystem, a single compromised or malfunctioning vehicle broadcasting erroneous safety data can lead to catastrophic real-world collisions. This rapid market expansion and deepening safety dependency underscore a critical requirement: maintaining overall system integrity relies on security architectures capable of swiftly identifying and revoking untrustworthy actors from the network.

1.1 Motivation

Vehicle-to-Everything (V2X) communication is being deployed specifically because it is safety-relevant: Cooperative Awareness Messages (CAM) and Decentralized Environmental Notification Messages (DENM) enable vehicles to exchange time-sensitive hazard alerts rapidly enough to directly influence automated braking and collision-avoidance systems. This safety dependency is exactly what makes the underlying Public Key Infrastructure (PKI) a hard requirement rather than a defensive add-on: a vehicle’s automated systems cannot act on a warning message without some assurance that the message is authentic, and they cannot verify authenticity without a functioning certificate infrastructure sitting underneath every safety message on the road.

Revocation is the part of that infrastructure most directly exposed to an adversary, and the part most often evaluated as an afterthought. Once a vehicle is known to be compromised, misbehaving, or otherwise untrustworthy, the certificate infrastructure’s only tool for removing it from the network is revocation — and until revocation takes effect, the vehicle’s messages remain cryptographically indistinguishable from a legitimate participant’s.

The published literature on V2X revocation, has overwhelmingly measured revocation schemes on a single axis: how fast do they work under ordinary network conditions? While baseline propagation speed is a necessary metric, it fails to capture the perspective of an active adversary. Rather than acting as a source of background packet loss, a scheme-aware attacker actively exploits the specific structural mechanics of the target revocation

protocol to maximize the temporal window during which their revoked credentials remain accepted. Evaluating true protocol resilience therefore demands modeling an adversary who strategically disrupts the enforcement pipeline. This fundamental distinction—between generic network unreliability and a deliberate, scheme-aware attacker—defines the core motivation of this thesis, establishing the analytical axis along which prior work on this codebase is extended.

1.2 Problem Statement

Evaluating V2X revocation protocols exclusively under benign network conditions creates a dangerous false equivalence: schemes that perform identically during normal operation may degrade in fundamentally different ways when subjected to targeted adversarial attack.

Prior work by Dehaes [7] and Orban [17] established the foundation for this research by implementing three major revocation paradigms—active, passive, and self-revocation—side by side on the Artery/SUMO/OMNeT++ simulation stack. While their work produced the first comparative evaluation of these three schemes on a common testbed, their scope was strictly limited to benign network conditions. By applying probabilistic packet loss and latency uniformly at the Registration Authority (RA) and wireless layers, their model approximated generic network unreliability rather than adversarial behavior directed at the revocation pipeline itself. This gap is pervasive across the broader literature. Existing adversarial studies, such as Scopelliti et al. [18], evaluate scheme-aware attackers strictly against a single scheme in isolation (self-revocation). Conversely, comparative surveys like Angelogianni et al. [2] evaluate multiple architectures, but only under nominal operating conditions. To date, no study in the published literature has evaluated active, passive, and self-revocation simultaneously under a unified, deliberately adversarial threat model.

This thesis addresses that gap directly. It evaluates how each of the three revocation schemes degrades when the packet loss or delay a vehicle experiences is no longer a random network condition but a strategy chosen by an attacker who specifically wants to evade or delay its own revocation, and — the comparative question the existing literature cannot answer — which scheme remains most robust once that shift is made. Efficiency under benign conditions and robustness under attack are distinct properties. Indeed, Dehaes and Orban’s finding that active and self-revocation achieve near-identical revocation times under normal conditions highlights why this adversarial comparison is essential, two systems that appear equivalent in a benign environment may exhibit starkly different failure modes once actively targeted.

1.3 Research Questions

- RQ1.** Under attacks that target each scheme’s own enforcement mechanism—rather than generic network unreliability—how do active, passive, and self-revocation compare in effective revocation time, message overhead, proportion of vehicles correctly revoked within a fixed time bound, and false-negative exposure window?
- RQ2.** Which structural properties of a scheme’s enforcement mechanism—push-based broadcast dependency versus pull-based local enforcement, explicit peer-side verification versus implicit desynchronisation, and central-authority trust boundary—explain differences in robustness observed under RQ1, as opposed to differences in

raw revocation speed under benign conditions?

1.4 Contributions

This thesis makes two primary contributions, each positioned directly relative to the specific prior work it extends:

1. **A Comparative Adversarial Evaluation on a Common Testbed:**

Where prior work by Dehaes and Orban evaluated the three revocation schemes (active, passive, and self-revocation) strictly under benign conditions, and Scopelliti et al. evaluated an attacker against only a single scheme in isolation, this thesis unifies both dimensions. Holding the simulation stack, traffic topology, and operational conditions constant, it subjects all three revocation schemes to a unified, multi-scheme adversarial evaluation on a shared testbed.

2. **Threat Modeling Grounded in Published Protocol Specifications:**

Rather than relying on generic, stochastic network degradation (such as uniform packet loss or random delay), the attack catalog defined in Chapter 3 is grounded strictly in the published protocol specifications, formalizations, and standards governing each scheme. Every evaluated attack targets a specific, named architectural property of how a given scheme processes and enforces revocation—such as self-revocation’s fundamental reliance on timestamp synchronization and un-replayed heartbeats—ensuring that the evaluation measures true protocol-level resilience against scheme-aware adversaries.

1.5 Non-Goals

The following are explicitly out of scope, stated here so that their absence later in the thesis is read as a deliberate scoping decision, not an omission:

- **Proposing a new revocation scheme.** This thesis evaluates the robustness of the three existing schemes as implemented; it does not design a fourth scheme, a hybrid scheme, or a modification intended to outperform the three under test.
- **Full ETSI TS 102 941 compliance.** The codebase inherited from Dehaes and Orban makes deliberate simplifications relative to a full ETSI-compliant deployment — a single RSU, single-hop broadcast with no peer relay, and a simplified pseudonym lifecycle, as detailed in Chapter 2. This thesis does not extend the codebase toward full standards compliance; where a simplification materially affects an attack’s external validity, this is flagged explicitly in Chapter 3 rather than corrected.
- **Switching simulation stack.** All experimental work is conducted on Artery/SUMO/OMNeT++, the stack used by both prior theses. No comparison to, or migration toward, an alternative simulator (e.g. Veins, ns-3) is undertaken. This also fixes the radio access technology at ITS-G5, as implemented by Artery’s underlying Vanetza library; comparison against the competing C-V2X radio access technology is out of scope for the same reason (Chapter 2, Section 2.1.1) — the revocation schemes under study operate above this layer and are not expected to depend on which one carries the traffic.

1.6 Thesis Structure

The remainder of this thesis follows the Plan–Do–Check–Act structure set out in the thesis template.

Chapter 2, State of the Art, surveys V2X communication and PKI fundamentals, describes the three revocation schemes and their prior evaluation, and reviews the existing attack literature on V2X revocation and credential management, concluding with the literature gap this thesis addresses.

Chapter 3, Threat Model and Attack Selection, formalises the attacker model motivated in Section 1.1 and grounded in Chapter 2, defines the six-criteria methodology used to select attacks, and presents the resulting attack catalog together with the candidates excluded and the rationale for excluding them.

Chapter 4, Implementation & Testing, establishes the methodology and empirical baseline the remainder of the thesis builds on: it defines how each comparison metric is computed reproduces Dehaes and Orban’s published benign-condition results for all three schemes as a consistency check, and documents how each selected attack is implemented, together with the calibration and justification behind every parameter swept in the experiments that follow.

Chapter 5, Experimentation & Data Collection, reports the results of each selected attack (A1, A5, A12) against the three revocation schemes: a headline comparison per attack placing all three schemes on a shared axis, followed by the per-scheme mechanism detail and comparative reading that explains the observed differences architecturally.

Chapter 6, Analysis & Discussion, interprets the results reported in Chapter 5: it answers RQ1 quantitatively and RQ2 architecturally, compares the findings against the state of the art surveyed in Chapter 2, and discusses the lessons learned and the limitations of the methodology and its results.

??, Conclusions, summarises the thesis’s answers to RQ1 and RQ2, states its limitations, and identifies directions for future work that this thesis’s own non-goals deliberately leave open.

Chapter 2

State of the Art

This chapter surveys the technical and academic background required to situate the thesis’s contribution. Section 2.1 introduces the V2X communication and public-key infrastructure (PKI) context in which certificate revocation operates. Section 2.2 describes the three revocation schemes evaluated in this thesis, their design goals, and the prior evaluation performed by Dehaes [7] and Orban [17] on the same simulator this thesis extends. Section 2.3 surveys the existing attack literature relevant to V2X revocation and credential management. Section 2.4 synthesises these three bodies of work into the specific gap this thesis addresses, which is then operationalised into a threat model and attack selection methodology in Chapter 3.

2.1 V2X Communication and Security Fundamentals

2.1.1 Safety-critical messaging and the case for a PKI

Vehicle-to-Everything (V2X) communication enables vehicles, roadside infrastructure, and vulnerable road users to continuously exchange status and hazard information over wireless links. In the European ETSI ITS-G5 stack, this is primarily realized through Cooperative Awareness Messages (CAMs) and Decentralized Environmental Notification Messages (DENMs). ITS-G5 represents the European profile of the IEEE 802.11p radio access technology standardized in North America as Dedicated Short-Range Communications (DSRC) [21]. The principal alternative, 3GPP Cellular V2X (C-V2X), carries V2X traffic over LTE/5G sidelink or cellular infrastructure [21]. Because the revocation mechanisms evaluated in this thesis operate strictly at the application and certificate layers, they are inherently agnostic to the underlying access technology. This work specifically builds upon Artery’s ITS-G5/Vanetza stack, treating access-layer transitions as out of scope.

Because safety-critical messages directly inform automated driving decisions—such as forward collision warnings, intersection assistance, and emergency braking—securing them is paramount. Because CAMs and DENMs are broadcast unencrypted to minimize processing latency, security does not rely on confidentiality, but rather on strict authenticity and integrity. A forged or tampered message presents an immediate physical hazard: an attacker capable of injecting ghost CAMs can induce phantom braking or dangerous evasive maneuvers in surrounding automated vehicles. To guarantee message origin and content integrity, modern V2X security architectures deploy a Public Key Infrastructure (PKI), requiring every outgoing message to be digitally signed and verified against a certificate chain rooted in a trusted authority.

However, a PKI alone, creates a new problem that is central to this thesis. If every vehicle used a single long-lived certificate to sign its messages, any receiver could trivially link successive messages to the same certificate and track that vehicle’s movements over time — a direct privacy violation. To mitigate this tracking risk, V2X PKI standards issue *pseudonym certificates*: short-lived, frequently rotated credentials that authenticate messages without revealing a stable, long-term identity. This architectural requirement renders certificate revocation in V2X significantly more complex than in classical settings

like TLS. Rather than invalidating a single static credential, a revocation mechanism must invalidate an entire stream of current and pre-issued pseudonyms cached by a misbehaving vehicle, all while preserving the privacy guarantees of non-revoked peers.

2.1.2 ETSI C-ITS and SCMS Security Architectures

In Europe, the security architecture for Cooperative Intelligent Transport Systems (C-ITS) is standardized by ETSI, primarily in TS 102 940 [10] (overall architecture) and TS 102 941 [11] (trust and privacy management). To balance message authenticity with driver privacy, the architecture separates credential issuance between two distinct authority entities:

Enrolment Authority (EA): Authenticates an ITS Station (ITS-S)—such as an On-Board Unit (OBU) in a vehicle or a Roadside Unit (RSU)—against its canonical, long-term identity and issues a long-term *enrolment credential*. This provisioning step occurs once or infrequently, analogous to device enrolment in classical PKIs.

Authorization Authority (AA): Processes requests backed by a valid enrolment credential and issues short-lived *Authorization Tickets (ATs)*—the ETSI terminology for pseudonym certificates. Day-to-day V2V and V2I communications are signed exclusively using ATs rather than long-term enrolment credentials.

A Root Certificate Authority (RCA) anchors trust for both the EA and the AA. Critically for revocation, the EA and AA are functionally and, in a full deployment, organisationally separate: the AA issuing pseudonyms remains unaware of the vehicle’s long-term identity, while the EA knowing the long-term identity cannot observe which pseudonyms were derived from it. This organizational decoupling enables ETSI’s passive revocation strategy (Section 2.2.2) to function without requiring a central authority to compile or maintain a master list linking active pseudonyms to real-world identities.

In North America, the Security Credential Management System (SCMS)—standardized under IEEE 1609.2 and detailed by Brecht et al. [6]—follows an analogous enrolment and authorization division. However, SCMS enforces an even stricter separation of duties by splitting pseudonym generation and linkage roles across independent Linkage Authorities, ensuring no single entity can unilaterally deanonymize a vehicle. This architectural choice directly underpins SCMS’s active revocation model (Section 2.2.1).

While ETSI and SCMS represent the two primary architectural paradigms in the V2X literature, the simulation framework utilized in this thesis models entity roles after the ETSI specification. Following the established implementation baseline of Dehaes and Orban, the EA, AA, and Revocation Authority (RA) functions are abstractly consolidated into a unified authority instance per scheme—a standard abstraction for V2X protocol simulation testbeds that isolates revocation performance without altering protocol-level messaging dynamics.

2.1.3 Revocation as a security primitive

To rigorously scope revocation within V2X security architectures, Scopelliti et al. [18] distinguish between two fundamental revocation triggers: *identity-based revocation* and *message-based revocation*. Identity-based revocation excludes a vehicle from the network based on its canonical, long-term identity—typically resulting from administrative policy

violations, private key compromise, or vendor-wide vulnerability disclosures. Conversely, message-based revocation excludes a vehicle based on the content of its peer-to-peer traffic, as adjudicated by a misbehaviour detection system (MDS) that reports suspect messages to the revocation authority. Both trigger the same downstream mechanism: once a decision to revoke is made, the vehicle’s long-term credential is blacklisted (denying future re-enrolment) and its currently-valid pseudonyms must be made unusable across the network before other vehicles will accept messages signed with them.

Accordingly, the scope of this thesis is delimited strictly to the *downstream enforcement and propagation pipeline* following an authenticated revocation decision. This work evaluates how revocation directives propagate, how rapidly peer verifiers achieve convergence, and how adversarial strategies degrade this enforcement process. It does not implement or evaluate an MDS; the initial trigger for revocation is modeled as an exogenous, already-authenticated event, consistent with the system boundaries established in Section 2.3.5 and formalized in Chapter 3.

2.2 Certificate Revocation Schemes

Three revocation strategies dominate the V2X literature and are implemented in the codebase this thesis extends: *active*, *passive*, and *self*-revocation. Table 2.1 summarises their enforcement mechanism and the party responsible for enforcing it; the following subsections describe each in turn.

Table 2.1: Enforcement mechanism and responsible party for the three revocation schemes under evaluation.

Scheme	Enforcement mechanism	Enforced by	Distribution
Active	Certificate Revocation List (CRL) check	Every receiving vehicle	Push (broadcast)
Passive	Pseudonym expiry + enrolment denial	RA/AA (at re-issuance time)	Pull (no broadcast)
Self	Local credential deletion	Revoked vehicle’s Trusted Component	Push (broadcast)

2.2.1 Active revocation

Active revocation, the strategy adopted by SCMS [6], represents the most direct adaptation of the classical X.509 CRL model to V2X. When the RA decides to revoke a pseudonym, it adds an identifier for that pseudonym — a hash of the public key, in the schemes surveyed here, rather than the pseudonym itself, to limit unnecessary exposure — to a CRL, which is then distributed to all vehicles, typically via RSU broadcast. Every vehicle checks every incoming V2V message’s signing certificate against its locally cached CRL before accepting the message.

The primary advantage of active revocation is immediacy: once a vehicle receives an updated CRL, enforcement is instantaneous and requires no cooperation from the revoked vehicle. However, its major limitation lies in list dissemination, as highlighted by Yoshizawa et al. [21] and documented in prior studies on this codebase [7, 17]. As the number of revoked pseudonyms scales, the CRL size expands proportionally. Delivering these growing structures to mobile entities under severe bandwidth limitations and dense traffic conditions poses a formidable network distribution challenge. As Yoshizawa et al. emphasize, a CRL update yields little security value if it fails to reach target verifiers within its operational deadline. This vulnerability window constitutes the exact operational property exploited

by the attack vectors evaluated in this thesis (e.g., CRL flooding, selective forwarding, and coverage-gap attacks; Chapter 3).

To mitigate this dissemination bottleneck, Khodaei and Papadimitratos [16] propose a vehicle-centric distribution framework that partitions CRL updates by geographic region and route, utilizing Bloom-filter fingerprints for efficient validation. While their model benchmarks this optimized redesign against a baseline incorporating both RSU broadcasts and epidemic peer-to-peer relays, the active-revocation architecture evaluated in this thesis represents a more constrained setup: a single-RSU, single-hop broadcast topology without peer-to-peer forwarding. The security implications of this architectural baseline are analyzed further in Section 2.3.2.

2.2.2 Passive revocation

Passive revocation, ETSI’s chosen strategy [11], avoids distributing any revocation-specific message at all. Pseudonym certificates are issued with a short, fixed validity period; a vehicle must periodically return to the AA to obtain fresh pseudonyms. When the RA decides to revoke a vehicle, it does not touch already-issued pseudonyms — it blacklists the vehicle’s long-term enrolment credential internally and simply declines the vehicle’s next re-authorization request. ETSI TS 102 941 §6.1.4 formalises this as the canonical definition of passive revocation: enforcement happens exclusively at the reload point, not at the receiving peer, which is why neither active revocation’s CRL nor self-revocation’s heartbeat has any counterpart in the passive scheme.

The scheme’s principal advantage is architectural simplicity and reduced network overhead — no broadcast channel dedicated to revocation state exists at all, and consequently, as shown in Section 2.2.4 and argued further in Chapter 3, no broadcast-disruption attack has anything to disrupt. This structural property, not an oversight in this thesis’s attack selection, is why passive revocation appears as immune or partially resilient against several of the attacks catalogued in Chapter 3. The scheme’s well-documented weakness is the mirror image of its strength: a revoked vehicle retains full communication capability until its last valid pseudonym expires, a window bounded below by the chosen pseudonym lifetime. Shortening that lifetime reduces the window but increases re-authorization traffic proportionally, a trade-off both Dehaes and Orban discuss and that neither prior thesis nor this one attempts to resolve, since doing so would mean proposing a new scheme, which is explicitly out of scope for this thesis.

2.2.3 Self-revocation

Self-revocation departs from active and passive architectures by requiring the target vehicle’s internal hardware to actively participate in its own credential exclusion. The strategy was first proposed by Förster et al. as REWIRE [13]: each vehicle is assumed to contain a Trusted Component (TC) — conceptually a TPM or TEE — that independently manages the vehicle’s cryptographic material. When the RA decides to revoke a vehicle, it broadcasts an Order for Self-Revocation (OSR) containing the target pseudonym’s identifier; the target’s own TC recognises the identifier as its own and deletes its credentials, after which it can no longer produce valid signatures. Because resolution to a real-world identity is never required, the scheme preserves privacy while achieving what is, in principle, immediate revocation with no CRL distribution problem.

REWIRE’s original design depended on a second, parallel mechanism to bound the case

where a vehicle simply ignores the OSR: periodic RA-broadcast keep-alive messages, whose absence for an extended period would trigger automatic self-revocation on the assumption that the vehicle had lost connectivity or been compromised. As both prior theses note, this dependence on a locally trusted, synchronised time source is fragile, and — a gap neither the original REWIRE paper nor the surrounding literature resolved — nothing in the design prevents a vehicle from selectively dropping only the OSR while continuing to process keep-alives normally, defeating revocation while appearing perfectly healthy to the network.

Improved self-revocation scheme

Scopelliti et al. [18] redesign self-revocation around a single, unified mechanism that closes this gap: the *heartbeat* (HB). The RA periodically broadcasts an authenticated HB containing a *Pending Revocation List* (PRL) — the identifiers of all pseudonyms currently scheduled for revocation — and a timestamp t_{hb} . Vehicles relay every HB to their TC, which (i) verifies the RA’s signature over the PRL and timestamp, (ii) checks freshness, $t_{hb} \geq now - T_v$ for a configurable validity window T_v , and (iii) on success, advances its internal clock to $now := \max(now, t_{hb})$. If the vehicle’s own pseudonym identifier ever appears in a PRL it relayed and accepted, the TC deletes its credentials directly.

Because the same message that carries revocation state also carries the only source of time the TC trusts, a vehicle cannot selectively suppress the PRL entry that targets it without also refusing to synchronise its clock at all — and an unsynchronised TC eventually desynchronises past T_v and is treated as unreachable. This single design decision is what formally bounds the *effective revocation time*: the authors define it as the interval between the RA’s revocation decision and the last verified message signed by the revoked pseudonym, and prove it has a deterministic upper bound under their four design objectives — guaranteed revocation (O1), no reliance on a locally trusted time source (O2), scalability (O3), and configurability for varying network conditions and security requirements (O4) — even against an attacker that drops or delays HBs, which they evaluate directly using three attacker sophistication tiers (*honest*, *smart*, *blind*), returned to in Section 2.3.3. The TC itself is assumed fully trusted and outside the attacker’s reach; only the untrusted OBU software that relays messages to it is considered compromised.

This design is the one implemented in the codebase this thesis extends. The thesis’s own code inspection (documented in Chapter 3) confirms two properties of the implementation worth flagging here because they recur throughout the attack catalog: enforcement at a *peer* vehicle operates exclusively through the same timestamp-desynchronisation logic used for self-enforcement — there is no separate check of an incoming sender’s certificate against a locally cached PRL, matching Scopelliti et al.’s own formalisation of peer-side enforcement rather than reflecting a simulator simplification — and HB signature verification in this implementation uses a key embedded in the HB message itself rather than a pre-provisioned RA certificate, a genuine simulator simplification rather than a protocol flaw, but one with direct consequences for the trusted-time attacks discussed in Section 2.3.3.

2.2.4 Prior comparative evaluation

All three schemes above were implemented side by side, in three otherwise comparable simulation scenarios, by Dehaes [7] and Orban [17] on the Artery/SUMO/OMNeT++ stack this thesis continues to use. Their joint evaluation, restricted to benign network conditions

— random packet loss and delay applied uniformly at the RA and radio layers, with no adversarial vehicle behaviour — is the direct empirical predecessor of this thesis and its baseline for effective revocation time and network overhead. Their central finding is that active and self-revocation achieve statistically similar median effective revocation times, both propagating revocation state at essentially the same rate. While passive revocation’s revocation time is, by construction, bounded below by the configured pseudonym lifetime and is therefore substantially higher across all tested parameter settings — an expected and design-intentional result rather than a deficiency, since the experiment set the pseudonym lifetime equal to the target revocation-time baseline.

Critically, neither thesis evaluates any condition in which a vehicle behaves adversarially with respect to the revocation mechanism itself. The packet loss and channel degradation modeled in prior work represent stochastic channel effects applied uniformly across all network traffic, rather than targeted evasion strategies deployed by a rational attacker who specifically wants to evade revocation. Reconfirming this baseline through preliminary codebase inspection formed the explicit motivation for the adversarial threat models introduced in this thesis.

To further contextualize the scope of this work within the broader empirical literature, Angelogianni et al. [2] independently compare a PKI-based architecture against a Direct Anonymous Attestation (DAA)-based one, including each architecture’s revocation capability, and measure signing/verification cost, end-to-end latency, pseudonym-reload efficiency, and revocation execution time across both. This is one of the few published attempts at a systematic, quantitative comparison across revocation strategies, and Orban explicitly flags it as the closest prior work to a comparative revocation framework in the literature. It differs from this thesis, and from Dehaes and Orban’s own comparison, along two axes that matter for how this thesis’s contribution should be framed: it compares architectures (PKI versus DAA), not revocation *schemes* within a single architecture, and — like Dehaes and Orban’s evaluation — it does not introduce any adversarial condition; all measurements are taken under normal operation. No source found in this literature review evaluates active, passive, and self-revocation together under a common, deliberately adversarial condition, a point developed fully in Section 2.4.

2.3 Existing Attack Literature on V2X Revocation and Credential Management

2.3.1 Attacker models and threat taxonomies

Two complementary frameworks recur throughout the literature reviewed for this thesis and are adopted directly in its own threat model (Section 3.1).

Yoshizawa et al. [21] classify V2X attackers along five independent dimensions — active/passive, local/remote scope, local/global view, insider/outsider, and individual/organised. This taxonomy provides a composable, fine-grained statement of attacker capabilities, avoiding monolithic or imprecise characterizations of adversarial strength.

Separately, formal-verification work on revocation protocols, including Scopelliti et al. [18] and Whitefield et al. [20], models the network itself as a Dolev-Yao adversary: an attacker who fully controls the network — able to intercept, reorder, delay, replay, and suppress any message — but who cannot break the underlying cryptographic primitives or compromise a vehicle’s TC. This is the dominant attacker model for revocation-specific

work because it isolates protocol-level weaknesses from cryptographic ones, and it is the model this thesis inherits for its own primary attacker (Section 3.1).

At the standards level, ETSI TR 102 893, the C-ITS threat, vulnerability, and risk analysis [8], defines a standardized threat, vulnerability, and risk taxonomy for C-ITS architectures. It establishes a structured threat-ID vocabulary covering message suppression, replay attacks, and denial-of-service (DoS) vectors targeting V2X infrastructure. Mapping the attack catalog in Chapter 3 directly to these ETSI threat identifiers ensures that the evaluated attack vectors align with established industry standards rather than ad hoc vulnerability descriptions.

2.3.2 Attacks on broadcast-based revocation distribution

Both active revocation’s CRL broadcast and self-revocation’s HB broadcast share a fundamental structural dependency on reliable message delivery, a vulnerability that the literature evaluates from two distinct operational directions: receiver-side suppression and channel-side degradation.

Selective, receiver-side suppression. Sedar et al. [19] characterise the *gray-hole* attack as attack as a scenario in which a compromised node routes standard network traffic normally while selectively dropping data packets belonging to a targeted application or service. Van der Heijden et al. [14] describe the same underlying pattern in more general terms, without using this specific label, as a routing-layer attacker that deviates from expected behaviour to cause specific messages to be lost rather than disrupting the network indiscriminately. This thesis adapts the pattern to a revocation-distribution channel: a compromised node continues to participate normally in all other V2X traffic, but selectively fails to process the specific messages — the CRL update or heartbeat — that would trigger its own exclusion, treating the revocation service itself as Sedar et al.’s targeted application. Because this class of attack targets message *processing at the receiver*, not the distribution channel itself, it applies differently to each scheme depending on where enforcement actually happens: it applies directly against active revocation (a vehicle simply never updates its local CRL) and self-revocation (a vehicle simply never relays HBs to its TC), but has no target at all against passive revocation, whose enforcement point is the RA’s re-authorization check, not anything the attacker’s own OBU processes.

Distribution-channel flooding and pollution. Khodaei and Papadimitratos [16] evaluate the opposite direction: rather than suppressing delivery to itself, a coalition of compromised, validly credentialed vehicles floods the CRL distribution channel with excess or fabricated CRL pieces, degrading delivery to *everyone*. Their evaluation methodology — tracking the percentage of “cognizant” vehicles (those holding an up-to-date CRL) over time as a function of attacker population share (0%, 1%, 5%, 10%, 25%, and 50% of vehicles) — is, to date, the most directly reusable experimental design found in this review for measuring a distribution-layer attack’s effect, and is adopted as a methodological template in Chapter 3. Their result that the baseline scheme degrades sharply once attackers exceed roughly 10% of the vehicle population, while their vehicle-centric scheme remains stable even at 50%, is directly relevant here — though both of their configurations use RSU broadcast and peer relay; the resilience gap comes from the vehicle-centric scheme’s trip-relevant filtering and free validation, not from relay presence. The active-revocation implementation evaluated in this thesis has no peer relay at all and no per-piece validation, operating under a more constrained topology than either configuration tested by Khodaei and Papadimitratos—a structural boundary explicitly accounted for in the empirical eval-

uation of Chapter 3.

2.3.3 Attacks on self-revocation and trusted time

Self-revocation’s dependence on the HB channel for both revocation state and time synchronisation makes it the most attack-studied of the three schemes in the literature, along two distinct lines.

Direct Evaluation of HB-Channel Attackers. Scopelliti et al. [18] evaluate self-revocation against three attacker sophistication tiers operating on HB delivery to the TC:

- i) an *honest* baseline that relays every received HB;
- ii) a *smart* attacker that relays standard HBs normally but selectively drops any HB containing its own pseudonym in the Pending Revocation List (PRL) while continuing to synchronize on untargeted HBs; and
- iii) a *blind* attacker (modeling encrypted HB content) forced to guess an optimal delivery strategy—specifically, relaying exactly one HB per validity window T_v to prevent desynchronization while minimizing exposure to revocation updates.

Significantly, even against the smart attacker, effective revocation time remains bounded because honest peer clocks advance past $t_{\text{rev}} + T_v$ regardless of the adversary’s actions. This finding is directly relevant to this thesis for two reasons: it serves as the direct theoretical foundation for the PRL-suppression attack evaluated in Chapter 3, and because Scopelliti et al. evaluated this strategy strictly in isolation within self-revocation, their analysis leaves open how an equivalent-capability adversary impacts active or passive revocation under the same topology—a comparative question addressed directly in this work.

Formally identified protocol flaws. Whitefield et al. [20] apply the Tamarin prover to REWIRE-family self-revocation protocols and identify two distinct flaw classes. In the “plain” protocol variant, a vehicle that changes its pseudonym before receiving an OSR referencing its old pseudonym silently ignores the message — a functional correctness gap rather than an active attack, since it requires no adversarial behaviour, only unlucky timing. A gap Förster et al.’s own original REWIRE paper had already flagged, which Whitefield et al.’s formal analysis independently reconfirmed. In the “R-token” variant, they identify a genuine authentication flaw: the confirmation message a vehicle sends back to the RA to acknowledge self-revocation can be forged by any adversary who observes the broadcast OSR, because the RA has no pre-established key against which to verify that the confirmation actually originated from the targeted vehicle. The authors’ proposed fix, the “O-token” scheme, binds a dedicated, RA-verifiable key pair to each pseudonym specifically for revocation confirmation. Neither flaw transfers directly to the HB/PRL design evaluated in this thesis, which has no OSR/confirmation exchange at all. This thesis treats Whitefield et al. as formal precedent establishing that this class of flaw is a recognised risk in this protocol family.

Trusted time as an open problem. A third line of work, concerns whether a TC can be assumed to have access to reliable time at all. Alder, Scopelliti, Van Bulck, and Mühlberg [1] characterise temporal guarantees in Trusted Execution Environments as an unsolved problem: under a realistic attacker model, a TEE cannot, in general, be assumed to have access to a trustworthy local clock, which is precisely the constraint self-revocation’s HB-based synchronisation (objective O2, Section 2.2.3) was designed around rather than assumed away. That this is a genuine, actively researched problem rather than a contrived

simulator artefact is confirmed by the wider trusted-time literature: Bettinger et al. [5] document that a prior, purpose-built distributed TEE time protocol (Triad) was vulnerable to exactly this class of attack — a single compromised node manipulating its own perceived time by delaying protocol messages to a remote time authority, an effect that could then cascade to honest peers — and design their own protocol, TriHaRd, specifically to close this gap, showing empirically that it detects and rejects this delay-based manipulation where the prior protocol could not. This is evidence that the underlying attack class is real and non-trivial to defend against, not that it remains open in the current state of the art. Anwar and Srivastava [3] complement this with a platform-level view, formalising three categories of timing attack against trusted execution environments: OS-level scheduling attacks, OS-level delay attacks on time packets, and network-level attacks against time synchronisation traffic. Collectively, this research confirms that HB delivery-delay—an attack requiring no cryptographic forgery or RA compromise—is not an artifact of simulation design, but rather the direct operationalisation of an open research problem in trusted computing, which is the framing this thesis adopts for its own trusted-time attack in Chapter 3.

2.3.4 Attacks on enrolment and credential re-issuance

A revocation scheme that correctly invalidates a vehicle’s current pseudonyms but allows the same vehicle to trivially re-obtain new ones has not actually revoked anything. Scopelliti et al.’s Requirement 6 [18] states this explicitly: only non-revoked vehicles may successfully re-enrol. ETSI TS 102 941 §6.2.3.2 [11] operationalises this requirement cryptographically, mandating that a re-enrolment or re-authorization request be authenticated by proof of possession of the vehicle’s current, valid credential — not merely a self-asserted identifier — which the associated Certificate Policy for European C-ITS [12] reinforces by treating re-keying as subject to the same authentication requirements as initial issuance. This is a narrow but consequential requirement: it is the only point in the surveyed literature where active and self-revocation’s push-based enforcement and passive revocation’s pull-based enforcement converge on an identical precondition. If it fails, it fails identically for all three schemes, regardless of how well each scheme otherwise propagates revocation state.

2.3.5 Misbehaviour Detection Systems and Scope Boundaries

A closely related family of attacks — Sybil attacks and false-accusation (or “bad-mouthing”) attacks against a misbehaviour detection system (MDS) — recurs prominently throughout the broader V2X security literature. These attack vectors target the *decision* to revoke: an attacker either creates multiple false identities to disproportionately influence a reputation or voting mechanism, or submits fabricated misbehaviour reports to frame an honest vehicle for revocation. Yoshizawa et al. [21] identify MDS design, and in particular plausibility validation for misbehaviour reports, as an open problem largely left to individual implementations by the ETSI standards, resulting in inconsistent coverage across deployments; ETSI TR 103 460 [9] explicitly flags Sybil-capable message-based attacks against detection systems as an unresolved challenge rather than a solved requirement for current releases.

As established in Section 2.1.3, this thesis strictly evaluates downstream propagation and enforcement mechanisms operating *after* a revocation decision has been rendered. Consequently, upstream MDS pipelines, along with Sybil and false-accusations attack vectors, fall outside the experimental scope of this work. Developing a standalone detection

and reporting framework—requiring novel telemetry message types, vehicle-side plausibility logic, and authority-side report processing—would represent an independent research effort. Bounding the scope strictly to downstream dissemination aligns with the standards gaps identified by Yoshizawa et al. and guarantees a controlled, comparative evaluation of active, passive, and self-revocation mechanisms.

2.4 The Gap This Thesis Addresses

Three distinct bodies of prior work bear directly on this thesis, and none of them, individually or combined, answers the question this thesis asks.

First, the only existing *comparative* evaluation of active, passive, and self-revocation on a common simulation stack — Dehaes and Orban’s prior work on this exact Artery codebase [7, 17] — is restricted entirely to benign network conditions. Their packet loss and delay model is applied uniformly, representing ordinary network unreliability, not a rational actor choosing when and what to drop in order to specifically evade revocation. Their own code, confirmed by this thesis’s independent inspection (??), contains no vehicle-side adversarial behaviour of any kind.

Second, the only body of work that evaluates a V2X revocation scheme under a genuinely adversarial, revocation-aware attacker — Scopelliti et al.’s own evaluation of their self-revocation design [18] with honest/smart/blind HB attackers — is confined to a single scheme, evaluated in isolation. It establishes that self-revocation is robust against a specific attacker capability, but offers no basis for comparison, because active and passive revocation were never subjected to an attacker of equivalent capability, on the same topology, in the same study. Whether active revocation’s CRL push or passive revocation’s expiry-based pull would degrade more, less, or differently under a comparably sophisticated attacker is a question the source literature does not answer, because it was never asked of more than one scheme at a time.

Third, the closest attempt at a genuinely cross-scheme, quantitative comparison — Angelogianni et al.’s evaluation of PKI versus DAA-based architectures [2] — compares architectural families under normal operating conditions, not revocation schemes under attack, and Orban’s own assessment of that work notes the absence of any open, reproducible framework that would let later researchers extend the comparison to additional schemes or adversarial conditions. This thesis’s own contribution can be read, in part, as beginning to answer that call, restricted specifically to revocation and to a deliberately adversarial setting.

Consequently, a central question remains unaddressed across the state of the art: Under a deliberate, scheme-aware adversary targeting specific downstream enforcement logic rather than generic channel reliability, which revocation scheme demonstrates the greatest resilience, and across which performance dimensions?

Addressing this research gap requires three unified capabilities that no single prior study delivers:

- i) **A Standardized Simulation Baseline:** A common platform holding topology, mobility, traffic density, and underlying physical-layer parameters constant across all three schemes—a baseline provided by the extended Artery stack;
- ii) **Scheme-Aware Adversarial Vectors:** A catalog of attack models explicitly designed to target the unique enforcement and dissemination mechanisms of each

scheme, grounded in the theoretical literature surveyed in Section 2.3; and

- iii) **A Unified Comparative Benchmark:** A consistent evaluation methodology to measure relative performance degradation across revocation delay, network overhead, and unrevoked exposure windows.

Formulating and evaluating these requirements constitutes the core focus of Chapter 3, which formally defines the attacker model this chapter has grounded in the literature and derives, from it and from direct inspection of the codebase, the specific attack catalog this thesis implements and evaluates.

Scope Boundaries and External Validity. To accurately delineate the external validity of this synthesis, certain structural parameters of the simulation stack must be explicitly noted. The experimental environment adopted in this work—matching the configuration of Dehaes and Orban—utilizes a single Roadside Unit (RSU), single-hop wireless broadcast without peer-to-peer relaying, and a simplified pseudonym lifecycle. The simulation framework incorporates controlled abstractions relative to a fully specified ETSI-compliant deployment. These topology choices represent deliberate experimental controls designed to isolate protocol-level enforcement behavior from complex routing artifacts. Where these structural choices impact the broader applicability or external validity of a specific attack vector, these boundaries are explicitly analyzed in the threat modeling of Chapter 3.

Chapter 3

Threat Model and Attack Selection

3.1 Threat Model

Defining a precise attacker model is a prerequisite for any rigorous adversarial evaluation: without establishing explicit boundaries on what an adversary can observe, control, and execute, the outcomes of individual attack experiments cannot be meaningfully compared or generalized. This thesis adopts the multidimensional taxonomy proposed by Yoshizawa et al. [21], which characterizes V2X adversaries along five orthogonal dimensions:

- i) **Active vs. Passive:** Active message manipulation versus passive channel eavesdropping;
- ii) **Local vs. Remote Scope:** Spatially bounded radio range versus network-wide reachability;
- iii) **Local vs. Global View:** Partial, local channel visibility versus complete network observation;
- iv) **Insider vs. Outsider:** Possession of valid, unrevoked system credentials versus unauthenticated status; and
- v) **Individual vs. Organized Capability:** A single autonomous actor versus coordinated multi-agent collusion.

Primary Attacker Classification. The primary adversary modeled throughout this work is classified as *active, local-scope, insider, and individual*: a single vehicle holding valid credentials that participates in the V2X network while behaving adverserially. This directly mirrors the compromised vehicle model formalized by Scopelliti et al. [18]—an attacker exercising root access over their own vehicle’s software stack with the primary objective of delaying or evading credential revocation for as long as possible.

Formal Capabilities and System Boundaries. Across all evaluated scenarios, the adversary is modeled as a *Dolev-Yao network adversary* [18]: they exercise full control over the communication channel—capable of intercepting, delaying, replaying, and suppressing arbitrary messages—but cannot break established cryptographic primitives. The hardware Trusted Component (TC) of honest vehicles is assumed to operate securely and correctly. Except in scenarios explicitly evaluating infrastructure compromise, the Revocation Authority is assumed to be fully trusted. Nation-state adversaries, coordinated multi-vehicle collusion, and simultaneous multi-entity infrastructure compromise are explicitly out of scope.

3.2 Attack Selection Methodology

3.2.1 Selection Criteria

To establish a rigorous comparative security benchmark, candidate attacks are evaluated against six defined criteria. These criteria ensure that the resulting experimental suite meaningfully differentiates how each revocation scheme degrades under adversarial pressure, permitting a clear robustness ranking across all three schemes. The criteria are divided into *classification metrics* (used to characterize attack properties) and *gating thresholds* (used for inclusion filtering).

Discriminative Power (Gating Threshold):

Measures the extent to which an attack produces measurably distinct operational outcomes across at least two of the three revocation schemes. An attack that degrades all three schemes identically provides no comparative insight and is assigned *Low* discriminative power. While an attack that affects all three schemes differently will score *High*.

Measurability (Gating Threshold):

Evaluates whether the attack’s impact can be quantified via precise, computable metrics within the simulation environment. This includes the baseline metrics inherited from Dehaes and Orban [7, 17]—effective revocation time and message overhead—supplemented by the proportion of stations successfully revoked and the false-negative exposure window.

Threat Model Coherence (Gating Threshold):

Assesses whether the required adversarial capabilities align strictly with the threat model formalized in Section 3.1 and remain grounded in established standards and literature.

Scheme Specificity:

Classifies whether the attack targets scheme-specific protocol logic (*Specific*) or shared underlying network infrastructure (*Shared*). Scheme-specific vectors provide deeper analytical value by revealing architectural vulnerabilities inherent to protocol design rather than physical topology.

Simulation Fidelity:

Gauges how accurately the simulated attack mechanism reflects its theoretical counterpart. Direct C++ implementation within service-layer components yields *High* fidelity, whereas modeling adversarial impact at higher abstraction levels (e.g., parameterizing elevated packet loss to represent jamming) yields *Medium* fidelity. Explicitly identifying these abstraction levels establishes boundaries for experimental validity.

Implementation Cost:

Reflects the engineering effort required to realize the attack within the Artery/OMNeT++ stack, categorized as configuration-level modification (*Cfg*), extension of existing service classes (*Ext*), or implementation of novel C++ modules and network descriptors (*New*).

Inclusion Decision Rule. To guarantee inclusion in the benchmark suite, a candidate attack must simultaneously achieve a rating of at least *Medium* across all three gating criteria: *Discriminative Power*, *Measurability*, and *Threat Model Coherence*. Candidate vectors that fail to satisfy any single threshold criterion are excluded from the experimental catalog, regardless of their scores across the remaining descriptive metrics.

3.2.2 Candidate Evaluation

A total of 12 candidate attack vectors were identified through a systematic review of the V2X security literature [14, 16, 18, 20, 21] and formal protocol analysis of the target Artery simulation codebase. Each candidate was systematically evaluated against the six selection criteria established in Section 3.2.1. Table 3.1 summarizes the scoring and inclusion decisions.

Notation used in the evaluation matrix is defined as follows:

- **Scores:** *H* = High, *M* = Medium, *L* = Low;
- **Specificity:** *Sp.* = Scheme-Specific protocol logic, *Sh.* = Shared infrastructure;
- **Implementation Cost:** *Cfg* = Configuration change, *Ext* = Extension of existing service class, *New* = Novel C++ module;
- **Decision Verdict:** ✓ = Included in benchmark suite, × = Excluded.

Table 3.1: Systematic evaluation of attack candidates against selection criteria. Threshold gating criteria (**Discriminative Power**, **Measurability**, **Threat Model Coherence**) are highlighted in bold.

ID	Candidate Attack Vector	Discrim.	Meas.	Coher.	Spec.	Fidel.	Cost	Verdict
A1	Selective Forwarding (Gray-Hole)	H	H	H	Sp.	H	Ext	✓
A5	Mass Revocation Scalability Stress	H	H	H	Sh.	H	Ext	✓
A12	Mobile Jamming Bubble	H	H	H	Sh.	M	Ext	✓
A2	Channel Jamming / Physical DoS	M	M	M	Sh.	L	Ext	×
A3	Compromised RSU / Infrastructure	L	L	M	Sh.	H	Ext	×
A4	GNSS / Location Spoofing	L	L	L	Sh.	L	Cfg	×
A6	Re-enrolment Bypass	M	H	H	Sp.	H	Ext	×
A7	Targeted HB/PRL suppression	M	H	H	Sp.	H	Ext	×
A8	Trusted-Time Manipulation (Delivery Delay)	M	H	H	Sp.	H	Ext	×
A9	Heartbeat Replay	M	H	H	Sp.	H	Ext	×
A10	Sybil / False Accusation	M	L	H	Sh.	L	New	×
A11	CRL flooding	M	H	H	Sp.	H	New	×

Selected Attack Catalog. The finalized experimental catalog comprises three core attack vectors (**A1**, **A5**, and **A12**), each satisfying all three gating criteria simultaneously while targeting distinct enforcement primitives:

- Selective Forwarding (A1):** Operationalizes selective receiver-side message suppression. Under active revocation, the adversary drops incoming CRL updates; under self-revocation, it suppresses Heartbeat (HB) and Pending Revocation List (PRL) relays to its local Trusted Component (TC). Folding targeted HB suppression into A1 establishes a unified selective forwarding benchmark across both active and self-revocation paradigms.
- Mass Revocation Scalability Stress (A5):** Evaluates downstream dissemination performance under elevated revocation volumes. Rather than modeling upstream misbehaviour detection logic, A5 injects a high volume of concurrent revocations to

stress active CRL payload sizes, self-revocation PRL processing limits and passive-revocation behavior.

- iii) **Mobile Jamming Bubble (A12):** Executes physical-layer denial of reception through localized, dynamic signal disruption. A designated subset of vehicles carries a fixed-radius interference envelope through the network topology, causing deterministic message loss for any receiver within range at the moment of transmission, regardless of scheme or message type. Restricting disruption strictly to the wireless medium isolates each scheme’s core dependency on reliable message delivery, independent of node-level protocol compliance or authority workload.

Exclusion Rationales. Candidate vectors failing one or more gating criteria, or falling outside the defined scope, were excluded from the benchmark:

- **Channel Jamming (A2):** Excluded, as originally scoped, due to low simulation fidelity (L). Representing physical-layer jamming via a flat, position-independent packet-loss parameter yields limited scheme-specific insight, and stochastic channel degradation of that kind was already comprehensively evaluated by Dehaes and Orban [7, 17]. The underlying threat is not abandoned, however: **A12** (Table 3.1) implements the same physical-layer jamming concept at the fidelity this candidate lacked, by making message loss position- and radius-dependent rather than a network-wide constant, and is carried forward into the benchmark suite in A2’s place.
- **Compromised RSU / Infrastructure (A3):** Excluded due to low discriminative power (L). In the evaluated topology, the compromised RSU hosts the primary RA entity. Infrastructure compromise induces total catastrophic failure across all three schemes identically, yielding no comparative signal.
- **GNSS / Location Spoofing (A4):** Excluded due to low discriminative power (L) and architectural disconnect. Position data is handled exclusively within the SUMO mobility simulator layer and has no interface with the C++ revocation service classes in Artery (no geo-scoped CRL/PRL logic exists in the protocol implementations).
- **Re-enrolment Bypass (A6):** Excluded despite scoring above threshold across all three gating criteria (Table 3.1). Under the ETSI V2X security architecture, enrolment credentials are permanently bound to a vehicle’s hardware trust anchor during initial provisioning. Because our threat model (Section 3.1) assumes an adversary bounded by standard cryptographic primitives, a revoked station cannot forge or re-initiate enrolment. Simulating an enrolment bypass would violate foundational ETSI trust axioms rather than evaluate a scheme-level vulnerability.
- **Targeted HB Suppression (A7):** Excluded as a standalone vector because it represents a specific instance of selective forwarding, which is comprehensively evaluated within the broader operational scope of A1.
- **Trusted-Time Manipulation (Delivery Delay) (A8):** Excluded due to functional redundancy and scheme-specific scope. A8 exclusively affects self-revocation by artificially delaying HB delivery. Because delaying HBs either defers PRL enforcement or forces clock desynchronization, the vector reduces functionally to HB suppression, which is already comprehensively evaluated within A1.

- **Heartbeat Replay (A9):** Excluded due to redundancy. Protocol specifications enforce temporal freshness via signed UTC timestamps within HBs, allowing receivers to instantly reject replayed messages. If an adversary holds and re-injects a valid HB within its current time window, the vector reduces functionally to delivery delay (A8).
- **Upstream Sybil (A10) and CRL Flooding (A11):** Excluded due to scope boundaries (Section 2.3.5). Modeling upstream misbehaviour detection system (MDS) reporting or raw flooding is out of scope; their downstream systemic effect (high revocation volume) is directly evaluated via A5.

3.3 Included Attack Specifications

3.3.1 A1: Selective Forwarding.

- **Adversary Mechanism:** Attacker-controlled nodes selectively drop incoming revocation broadcasts (CRLs/PRLs) while continuing normal V2X communication (e.g., CAMs/DENMs).
- **Threat Model Scope:** Requires an insider adversary controlling $p \in \{10\%, 25\%, 50\%\}$ of the vehicle population.
- **Targeted Vulnerability:** Tests the dependency of active revocation on multi-hop broadcast propagation versus self-revocation’s local clock-desynchronization fallback.

3.3.2 A5: Mass Revocation Scalability Stress.

- **Adversary Mechanism:** Injection of spurious misbehaviour reports or manufactured false positives, driving elevated revocation decision volume at the Registration Authority (RA).
- **Threat Model Scope:** System-level stress test (0% vehicle-side attackers; operationalized as RA decision rate scaling).
- **Targeted Vulnerability:** Tests network overhead limits, payload fragmentation boundaries, and processing latency under heavy CRL/PRL sizes.

3.3.3 A12: Mobile Jamming Bubble.

- **Adversary Mechanism:** A designated subset of vehicles carry a fixed-radius zone of localized signal interference that travels with them through the network; any vehicle within that radius at the moment a message would arrive fails to receive it, regardless of scheme, sender, or message type.
- **Threat Model Scope:** Assumes a physical-layer adversary with mobility capabilities (e.g., an RF-jamming vehicle). System resilience is evaluated across four distinct configurations—Benign (0 m), Config A (25 m), Config B (100 m), and Config C (200 m)—to progressively scale the induced V2V censoring fraction.
- **Targeted Vulnerability:** Evaluates scheme reliance on uninhibited physical-layer delivery, testing how channel loss impacts active CRL propagation, self-revocation heartbeat synchronization, and passive re-enrollment availability.

Chapter 4

Implementation & Testing

4.1 Methodology

4.1.1 Testing methodology

The experimental procedure established in this chapter follows a strict test-driven evaluation model. Before introducing any adversarial actor defined in Chapter 3 into the simulator, this thesis verifies that the inherited codebase, executed under the benign conditions documented by its authors [7, 17], faithfully executes its documented design.

This verification is a methodological prerequisite. Every experimental result reported in Chapter 6 is quantified as a delta (Δ) between an adversarial run and a benign baseline. If the baseline configuration contains hidden parameter mismatches or incomplete logging, observed performance degradations under attack cannot be definitively attributed to the threat vector.

4.1.2 Definition of correctness adopted

Within this chapter, *correctness* is strictly defined as *conformance of the software implementation to the design documented in predecessor theses* [7, 17]. It does not denote complete adherence to ETSI TS 102 941 [11]—which remains outside the scope of this work—nor absolute cryptographic perfection. Inherited architectural simplifications are documented in ?? rather than modified. The evaluation addresses a specific question: *does the codebase execute as documented?* Any divergence identified during audit is documented as a structural finding and rectified to match the theoretical model.

4.1.3 Two-part verification procedure

Verification proceeds across two orthogonal axes:

(a) Parameter Alignment. The baseline configuration targets the setup specified by Orban [17]. The SUMO network topology consists of a closed-loop circuit featuring two vehicle insertion points, each spawning vehicles at 200 h^{-1} (400 h^{-1} combined), yielding 400 total vehicles over a 3600 s scenario with a steady-state road density of 30–35 active nodes. A single Roadside Unit (RSU) provides continuous omnidirectional coverage. Probabilistic channel impairments affect RSU-originated security management messages—specifically Certificate Revocation List (CRL) updates, pseudonyms, and self-revocation heartbeats—at a 40 % packet drop rate and an independent 40 % delay rate (normally distributed with $\mu = 2 \text{ s}$, $\sigma = 0.5 \text{ s}$) [17].

The self-revocation validity window is set to $T_v = 30 \text{ s}$. Heartbeat and CRL distribution intervals are calibrated to $T_v/5 = 6 \text{ s}$. Under a 40 % drop probability, five transmission attempts within one validity window bound total delivery failure to $0.4^5 \approx 1 \%$ [17]. Revocation decisions occur at uniform random intervals between 45 s and 85 s, revoking 1–3 vehicles per event (averaging 120 revoked vehicles per run).

(b) Instrumentation completeness. Logging and metrics-collection hooks were audited prior to benchmark execution. Each metric was validated for *completeness* (every event generates a log entry) and *soundness* (log entries correspond strictly to their target event type). Correcting instrumentation gaps was necessary to ensure that measurement noise was not mistaken for protocol behavior.

4.1.4 Metrics and how they are computed

The performance axes evaluated in Chapter 6 expand upon the metrics established by Dehaes [7]:

- **Effective Revocation Time:** The duration between the Registration Authority’s (RA) revocation decision for a target certificate and the *final* valid message accepted network-wide from that revoked certificate. Measuring first-rejection latency is insufficient, as non-uniform propagation allows un-updated nodes to continue accepting invalid messages. Computing this metric requires post-processing the global message acceptance log post-simulation. Right-censored instances (vehicles exiting the topology prior to final acceptance) are identified and isolated.
- **Protocol Overhead:** Measured via payload size expansion, packet delivery frequency, and aggregate byte consumption across simulation runs.
- **Enforcement Rate & False-Negative Exposure:** Formally introduced in this work, tracking the percentage of nodes enforcing revocation and quantifying total post-revocation accepted packets.

4.1.5 Analytic oracles

Verification outcomes are benchmarked against two theoretical reference bounds:

1. **Self-Revocation Hard Bound:** Theoretical self-revocation latency is strictly bounded above by $2T_v$ (60 s at $T_v = 30$ s) under benign conditions [18]. Any baseline observation exceeding 60 s indicates implementation defect or metric miscalculation.
2. **Predecessor Literature Consistency:** Measured metrics must align statistically with published figures from baseline studies [7, 17], with any variance traced to identified architectural fixes.

4.1.6 Relationship to Subsequent Chapters

The verified methodology and baseline configurations established in this chapter form the foundation for the adversarial experiments in Chapter 6. By eliminating instrumentation bias and confirming software alignment with ETSI-inspired design parameters, performance deltas measured under attacks ($A1$, $A5$, $A12$) can be directly attributed to protocol-level vulnerabilities rather than simulator artefacts.

4.2 Simulation Environment & Infrastructure Setup

To isolate adversarial impact from experimental noise, this evaluation reuses the Artery/SUMO/OMNeT++ substrate inherited directly from Dehaes and Orban [7, 17]. Keeping the software stack, topology, and parameter calibration unchanged ensures that

any performance degradation observed in Chapter 5 stems solely from attacker behavior rather than shifts in background traffic or channel loss. This setup directly adheres to Non-Goal 3 (Chapter 1): extending the established benign baseline rather than re-deriving it.

This section details the fixed substrate: the software stack (Section 4.2.1), the network topology and mobility model (Section 4.2.2), and the baseline parameter calibration alongside an empirical reproduction check (Section 4.2.3).

4.2.1 Software Stack

The simulation environment extends the Dehaes/Orban Artery’s fork, executed within the OMNeT++ discrete-event framework. The software stack follows a modular, layered architecture:

- **Application & Middleware Layer:** Artery manages V2X service lifecycles and message multiplexing. The three evaluated revocation mechanisms (active, passive, and self-revocation) are instantiated as independent simulation scenarios, each executing a single Revocation Authority (RA) service on `rsu[0]`.
- **Facilities & Mobility Layer:** The Facilities layer interfaces with SUMO via TraCI to dynamically synchronize vehicle positions and mobility states.
- **Networking & Security Layer:** Vanetza—integrated directly into Artery—handles the ETSI ITS-G5 protocol suite, including GeoNetworking routing, BTP, message security, and decentralised congestion control [7, 17].
- **Lower Layers:** Physical and link-layer radio propagation models are provided by the INET framework.

No modifications were made to this underlying infrastructure; all protocol extensions and adversarial behaviors introduced in this thesis are strictly localized to vehicle-side and RSU-side service logic (??).

4.2.2 Network Topology & Mobility Model

Vehicle mobility is simulated by SUMO on a closed-loop circuit generated in NetEdit, adopting the network layout established by Dehaes [7]. The baseline mobility configuration adheres to the following parameters:

- **Traffic Inflow:** Two vehicle spawn points operating at 200 vehicles/h each (400 vehicles/h total inflow), producing 400 unique vehicles per 1 h simulation run [7, 17].
- **Network Density & Speed:** A steady-state population of 30–35 concurrent vehicles moving at an average speed of 50 km h^{-1} .
- **Infrastructure Coverage:** A single RSU deployed to maintain continuous, omnidirectional coverage over the entire loop, ensuring unbroken connectivity with the Revocation Authority throughout a vehicle’s lifespan.

This full-coverage configuration serves as the benign baseline.

4.2.3 Parameter Calibration & Baseline Setup

To confirm that the inherited environment described in Sections 4.2.1 and 4.2.2 reproduces the behavior reported by Dehaes [7], baseline reproduction runs were executed for each scheme using the parameters specified in [7, Table 4.1]: a 6 s CRL generation interval for active revocation, a 6 s heartbeat interval with a 30 s validity window for self-revocation, and a 60 s pseudonym lifetime for passive revocation. All three runs applied identical channel impairment parameters to RSU-originated critical security messages: a 40 % packet drop probability and an independent 40 % delay probability drawn from a normal distribution ($\mu = 2.0\text{ s}, \sigma = 0.5\text{ s}$). Tables 4.1 and 4.2 compare the resulting effective revocation time and network overhead against Dehaes’ published baseline values.

Table 4.1: Effective revocation time, baseline parameters.

Scheme	(Median/Mean)	Dehaes [7]	This work	Δ
Active	Mean	10.65 s	7.81 s	−26.7%
Active	Median	8.78 s	5.43 s	−38.2%
Self	Mean	7.85 s	8.39 s	+6.9%
Self	Median	6.87 s	6.72 s	−2.2%
Passive	Mean	25.92 s	24.91 s	−3.9%
Passive	Median	24.50 s	24.06 s	−1.8%

Table 4.2: Network overhead, baseline parameters.

Scheme (total / avg / max message size)	Dehaes [7]	This work	Δ
Active — total CRL overhead	350.45 KiB	326.70 KiB	−6.8%
Active — avg / max CRL size	1025.30 B / 1552 B	937.09 B / 1364 B	−8.6% / −12.1%
Self — total HB overhead	175.15 KiB	177.02 KiB	+1.1%
Self — avg / max HB size	510.97 B / 544 B	514.98 B / 548 B	+0.8% / +0.7%
Passive — total combined overhead	3475.05 KiB	3981.22 KiB	+14.6%
Passive — avg / max message size	345.55 B / 488 B	487.07 B / 752 B	+41.0% / +54.1%

The empirical results for active and self-revocation faithfully reproduce the ordering and order of magnitude reported by Dehaes [7]: both remain single-digit-second mechanisms, roughly an order of magnitude faster than passive revocation. This outcome aligns with the structural expectation that proactive, push-based mechanisms (i.e., CRL broadcasts and heartbeat notifications) converge significantly faster than reactive approaches that enforce revocation only at a vehicle’s next re-enrollment boundary. Minor numerical discrepancies between our measurements and Dehaes’ published baselines are attributable to simulation stochasticity, including pseudo-random seed initialization, transient MAC-layer contention, and wireless channel realization noise.

4.3 Attack Implementation

4.3.1 Implementation Philosophy

Each attack is implemented to reproduce the functional consequence a real adversary would cause on the revocation process, rather than every low-level mechanism by which a real adversary might cause it. Where the target of the attack is a specific vehicle’s protocol behaviour (A1), a variant of that behaviour is introduced directly. Where the target is the operating conditions of the Authority issuing revocations (A5), only those conditions

are altered, without introducing any new vehicle-side logic. Where the target is the radio channel itself (A12), the propagation loss a jammer would cause is abstracted into a simple reception rule, consistent with the fidelity trade-off already justified in Chapter 3. In all cases, the honest baseline exists as a parameter boundary of the single underlying codebase (e.g., zero attacker share or zero jamming radius), preventing code drift and guaranteeing direct comparability between baseline and under-attack evaluations.

4.3.2 A1: Selective Forwarding

A1 models an insider adversary that behaves as a gray-hole attacker [19]: it forwards ordinary traffic normally while silently discarding one specific class of message. The objective is to test what happens to each revocation scheme once a fraction of the vehicle population simply refuses to receive the information that would let it enforce a revocation, without ever signalling to the rest of the network that anything is wrong.

The attack is implemented as a variant of the honest vehicle logic for each scheme under test. At the start of a run, a configurable share of the vehicle population is designated as attackers; every other vehicle behaves exactly as in the baseline scenario. An attacker vehicle continues to enroll, to broadcast and receive ordinary V2V traffic, and to behave indistinguishably from an honest vehicle on every channel except one: it discards, without processing, the specific message that would update its knowledge of revoked vehicles – the Certificate Revocation List broadcast under active revocation, or the heartbeat carrying the Pending Revocation List under self-revocation. Because only that one message class is affected, the attacker cannot be identified from its other communication patterns, matching the selective, non-disruptive character the attack is meant to represent rather than a blanket denial of service.

Self-revocation’s attacker retains one further piece of ordinary behaviour that turns out to matter for the results (Section 5.1.1): it still tracks its own internal clock exactly as an honest vehicle would. An attacker that stops processing heartbeats also stops advancing that clock, which independently triggers self-revocation’s desynchronisation safeguard regardless of whether it ever intended to comply. This was not designed into the attack; it follows naturally from implementing the attacker as a faithful variant of the honest vehicle rather than a hand-crafted adversary that only does what the attack strictly requires.

Exemption of Passive Revocation. Passive revocation is structurally excluded from evaluation under attack vector A1. Unlike active and self-revocation, passive revocation does not rely on broadcast updates to propagate revocation decisions across the V2V network. Instead, enforcement occurs deterministically at the Revocation Authority during pseudonym renewal: a revoked vehicle operates normally until its current short-lived certificate expires, after which its re-enrollment request is denied. Because there are no client-side broadcast updates for an adversary to selectively filter, selective forwarding cannot delay or bypass passive revocation enforcement.

Parameter Calibration and Justification. The attacker share is varied across four values: 0%, 10%, 25%, and 50% of the vehicle population. The zero-share case serves the same regression-verification role described above: it confirms the attacker mechanism, once present in the scenario configuration, has no effect when inactive. The three non-zero values are drawn from the attacker-population-share sweep design of Khodaei and

Papadimitratos [16], adopted as an evaluation template in Section 2.3.2, not the underlying attack.

4.3.3 A5: Mass Revocation Scalability Stress / Revocation Flooding

A5 represents a different kind of pressure on the system: not a non-compliant vehicle, but a surge of legitimate-looking revocation decisions, of the kind an upstream misbehaviour-detection system would produce if it generated false positives at an elevated rate. Modelling that detection process itself is outside the scope of this work; what matters for the schemes under test is only its downstream effect, a higher volume of revocations to distribute and enforce.

Accordingly, A5 introduces no vehicle-side attacker at all. It is implemented entirely as a change to the Authority’s own operating parameters: revocation decisions already occur at random intervals within a configured range, and A5 narrows that range so that more decisions occur over the same run duration. Everything else about how a revocation is decided, formatted, and disseminated is left untouched, so the attack varies revocation *volume* in isolation rather than introducing a second, confounding change to the schemes’ own logic that would make it harder to attribute an observed effect to volume specifically.

Sustaining several hundred revocations over a single run does, however, expose a capacity limit unrelated to the attack’s own design: certificate-revocation and pending-revocation lists grow with every entry, and the underlying network layer imposes a maximum message size. Active and self-revocation’s list capacity, together with the corresponding network-layer limit, were raised in step to accommodate the higher volumes tested, and this adjustment was validated separately as safe before being used (??). Passive revocation has no equivalent concern: because it enforces through a private, unicast blacklist rather than a growing message every vehicle must receive, its enforcement cost does not scale with revocation volume the way the other two schemes’ broadcast payloads do – one of the few respects in which passive revocation is structurally advantaged over the alternatives under this attack.

Parameter Calibration and Justification. The three revocation-volume levels evaluated – Baseline, ≈ 150 , and ≈ 200 – are reached by narrowing the interval from which the Authority draws the time to its next revocation decision, rather than by fixing an exact revocation count directly. The baseline configuration draws this interval uniformly from [45, 85] s (the unattacked default), which is narrowed to [32, 63] s for the ≈ 150 target and further to [25, 48] s for the ≈ 200 target, identically for both active and self-revocation. Because the resulting revocation count is a consequence of this interval rather than a value fixed directly, “ ≈ 150 ” and “ ≈ 200 ” denote calibrated targets rather than exact per-run counts, which vary with the same run-to-run stochasticity documented throughout this chapter’s baseline verification.

These two target volumes were chosen relative to two fixed reference points. At the low end, they sit clearly above the baseline’s own naturally-occurring revocation count (≈ 110 over a 3600 s run), so that the ≈ 150 configuration already represents an unambiguous increase rather than one indistinguishable from ordinary run-to-run variation. Passive revocation required its own, more aggressively narrowed intervals to reach comparable volumes, reflecting that it was calibrated independently rather than sharing active and self-revocation’s timing configuration.

4.3.4 A12: Mobile Jamming Bubble Implementation

A12 targets the transmission medium itself rather than any vehicle’s or Authority’s logic: a physical jammer that prevents nearby vehicles from receiving any radio message at all for as long as they remain within range.

Rather than simulating radio-layer signal propagation and interference directly, which the simulator’s abstraction level does not support with useful fidelity, the effect of jamming is modelled at the point where it ultimately matters: message reception. A subset of vehicles in the simulated fleet is designated as jammers; each carries an invisible zone of fixed radius around its current position, and any vehicle inside that zone at the moment a message would arrive fails to receive it, regardless of scheme or message type. Because jammers are ordinary vehicles following the same traffic simulation as everyone else, the affected zone moves with them rather than sitting at a fixed location – approximating a jamming source travelling with traffic (for instance, an adversarial vehicle) rather than a fixed installation, and exposing the same population of vehicles to jamming at different points in the network over the course of a run instead of only near one static location. The jamming radius and the time window during which it is active are both configurable, so the same mechanism supports evaluating the schemes under increasingly severe jamming conditions without changing anything about how jamming itself is modelled.

Parameter Calibration and Justification. The jamming radius (R_j) is varied across four configurations: Benign ($R_j = 0$ m), Config A (25 m), Config B (100 m), and Config C (200 m). The zero-radius case establishes the unattacked baseline, confirming that the jamming extension reduces to ordinary behaviour when disabled rather than introducing any systematic drift of its own. The three non-zero radii were not fixed from a nominal radio range alone; they were calibrated empirically against the resulting event-level V2V censoring fraction – the share of message-reception attempts a receiver never gets the chance to act on (Table 5.3) – so that the three configurations land at meaningfully different, well-separated severity levels rather than clustering together.

This calibration was itself an iterative process. An early, considerably larger radius was found to suppress essentially all V2V traffic across the simulated road loop, leaving no usable message-acceptance data for any scheme and nothing to compare against the unattacked baseline: a total communication blackout is uninformative for a comparative benchmark, since every scheme fails identically once no message can be delivered at all. The three radii retained here were chosen specifically to stop short of that failure mode: Config A represents a mild condition, censoring only 2.9–3.7% of V2V receptions across the three schemes; Config B a moderate condition, at 14.3–19.5%; and Config C the most severe condition evaluated, at 49.6–66.0% – substantial, but still leaving a majority of receptions unaffected in every scheme, so that each configuration continues to yield a large enough population of both censored and uncensored events for the enforcement-timing and false-positive metrics reported in Section 5.1.3 to remain statistically meaningful.

The chosen radii also correspond to plausible physical scales for a mobile jamming source – 25 m is on the order of an immediate single-hop V2V range, 100 m approaches typical urban/suburban ITS-G5 communication range, and 200 m approaches the outer bound of direct vehicular transmission range – but this correspondence is a secondary observation made after the fact, not the basis on which the three values were selected.

Chapter 5

Experimentation & data collection

5.1 Results

5.1.1 A1 — Results

Attack vector A1 was evaluated across four attacker population shares (0%, 10%, 25%, and 50%) for both active and self-revocation schemes; passive revocation is structurally excluded from this evaluation. All measurements utilize an updated metric evaluation counting any hash as a failure if at least one attacker-tagged vehicle evaded enforcement.

Table 5.1 presents the headline results comparing active and self-revocation across the evaluated attacker shares. The table records the median time-to-honest-enforcement and wrongful accepts for active revocation, alongside the same last-accepted-message effective revocation time (receiver-side, network-wide) and wrongful message receipts for self-revocation, computed the same way as active’s own metric.

Attacker share	Scheme	Median enforcement time	Wrongful accepts
0%	Active	3.33 s	8,380 (0.42%)
	Self	4.28 s	5,779 (0.31%)
10%	Active	5.88 s	24,707 (1.25%)
	Self	5.31 s	6,751 (0.47%)
25%	Active	5.15 s	59,211 (2.99%)
	Self	5.54 s	5,776 (0.55%)
50%	Active	4.91 s	72,329 (3.55%)
	Self	11.68 s	3,715 (0.63%)

Table 5.1: Headline performance comparison between active and self-revocation schemes under attack A1 across varying attacker population shares.

For active revocation, wrongful accepts represent genuine receiver-side propagation failures where an un-updated vehicle accepts a message from a revoked sender. For self-revocation, V2V message acceptance is governed by timestamp freshness rather than a local CRL check; consequently, the wrongful count reflects the revoked vehicle’s internal compliance delay between the RA decision and completion of its local PRL match or time desynchronization. This is not a fully independent measurement of network propagation the way it is for active, however: self-revocation performs no receiver-side revocation check at all, so the last accepted message from a given vehicle is, in practice, almost always bounded by that same vehicle’s own self-revocation completion time (it simply stops transmitting once revoked) rather than by how long it takes other receivers to converge.

Figure 5.1 details the distribution of revocation enforcement outcomes across attacker population shares for both schemes. Active revocation partitions outcomes into *enforced*, *evaded-by-attacker*, and *never-enforced*. The *evaded-by-attacker* category records any hash where an attacker vehicle accepted a post-revocation message, regardless of whether honest receivers enforced it.

Self-revocation partitions outcomes into *PRL-match*, *desync-fallback*, and *never-enforced*. The *desync-fallback* category captures instances where heartbeat disruption triggered a local time desynchronization, effecting functional revocation when the primary PRL mechanism was bypassed. Across both schemes, the *never-enforced* outcome accounts

for revoked vehicles that departed the simulation scenario prior to enforcement completion.

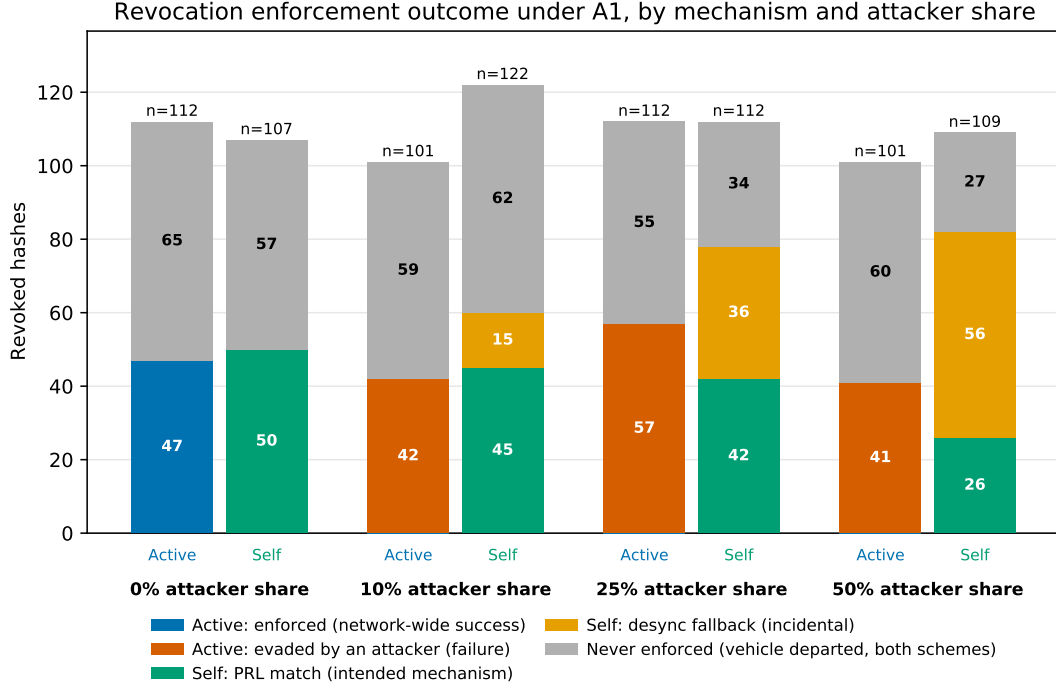


Figure 5.1: Breakdown of revocation enforcement outcomes by mechanism and attacker share for active and self-revocation schemes.

5.1.2 A5 — Results

Attack vector A5 evaluates system performance under elevated revocation decision volumes generated by false-positive misbehavior detections at the RA. Evaluation was conducted across three target revocation volume levels: Baseline (standard interval), ≈ 150 , and ≈ 200 total revocations per 3600 s run (with passive revocation evaluated at ≈ 100 , ≈ 150 , and ≈ 200). The attacker population share is fixed at 0% throughout.

Enforcement metrics and message acceptance behavior Table 5.2 provides a comparative summary of active, self, and passive revocation performance across the evaluated volume levels. Reported metrics include total revoked count (n), percentage of correctly enforced revocations, median enforcement time, and total post-revocation accepted messages (both raw count and percentage of network-wide volume).

Active revocation evaluates V2V acceptance against a local CRL, measuring genuine post-revocation propagation leaks. Self-revocation performs no V2V CRL check, meaning its post-revocation acceptances measure vehicle compliance latency. Passive revocation performs no V2V revocation check either, so its wrongful accepts measure something closer to active’s propagation-leak reading than self’s: every message from a still-validly-signed pseudonym is accepted by every peer regardless of RA revocation status, so the count reflects the volume of traffic a revoked vehicle’s current pseudonym generates during the window between the RA’s revocation decision and that pseudonym’s natural expiry or the vehicle’s next (denied) re-enrollment attempt, resolved via the vehicle-to-certificate-hash mapping recorded at issuance time (Table 4.1).

Scheme	Volume	Revoked (n)	% correct/enforced	Median enforcement time	Wrongful/leaked accepts
Active	Baseline	108	47.2%	5.43 s	10,922 (0.56%)
	≈ 150	146	51.4%	3.73 s	14,016 (0.74%)
	≈ 200	190	56.8%	6.00 s	21,844 (1.26%)
Self	Baseline	113	41.6%	6.59 s	7,583 (0.41%)
	≈ 150	154	57.8%	3.90 s	9,740 (0.60%)
	≈ 200	195	64.1%	6.40 s	16,261 (1.14%)
Passive	≈ 100	98	30.6%	36.62 s	8,741 (0.35%)
	≈ 150	158	31.6%	30.33 s	14,017 (0.58%)
	≈ 200	208	38.5%	26.84 s	19,726 (0.89%)

Table 5.2: Comparative summary of enforcement metrics and message acceptance behavior under attack A5 across active, self, and passive revocation schemes.

Bandwidth consumption versus revocation volume Figure 5.2 plots total revocation distribution bandwidth consumption in kilobytes over the full run against revocation volume on a logarithmic scale. Active and self-revocation totals represent aggregate CRL and Heartbeat broadcast transmissions multiplied by message size. Passive revocation bandwidth reflects total network traffic generated by pseudonym renewals and enrollment requests. Relative percentage changes from initial to maximum volume are annotated directly for each scheme.

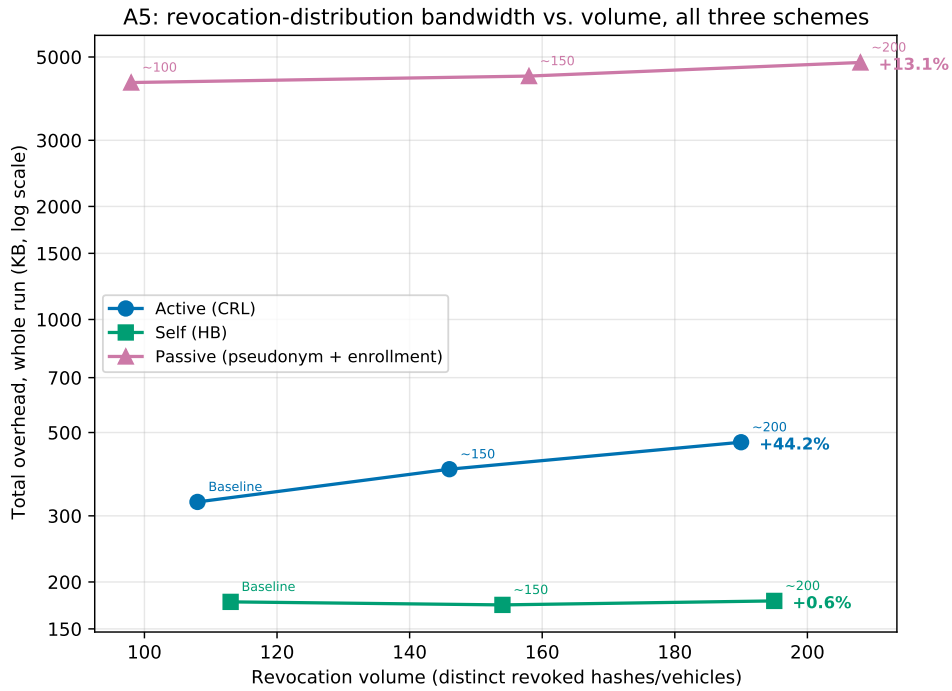


Figure 5.2: Total revocation distribution bandwidth consumption versus revocation volume across active, self, and passive revocation schemes.

5.1.3 A12 — Results

Attack vector A12 evaluates the operational resilience of all three schemes against localized RF jamming across four jammer radius configurations: Benign (0 m), Config A (25 m), Config B (100 m), and Config C (200 m).

Jamming severity per radius Table 5.3 details V2V message reception counts and event-level censoring fractions per scheme and configuration. The dropped message count is deduplicated per receiving vehicle reception event. The censoring fraction is defined as the ratio of dropped messages to total reception attempts (dropped/(dropped + not dropped)) and serves as the primary metric for jamming severity.

Scheme	Config (radius)	Messages dropped	Messages not dropped	Censoring fraction
Active	Benign	0	2,117,080	0.0%
	A (25 m)	67,979	1,873,995	3.5%
	B (100 m)	285,711	1,182,060	19.5%
	C (200 m)	609,957	314,318	66.0%
Self	Benign	0	1,846,872	0.0%
	A (25 m)	46,029	1,566,710	2.9%
	B (100 m)	159,895	956,799	14.3%
	C (200 m)	99,770	101,330	49.6%
Passive	Benign	0	1,325,407	0.0%
	A (25 m)	48,035	1,243,755	3.7%
	B (100 m)	225,125	1,041,894	17.8%
	C (200 m)	565,420	481,517	54.0%

Table 5.3: V2V message reception statistics and calculated event-level censoring fractions across jammer radius configurations for active, self, and passive schemes under attack A12.

Figure 5.3 displays median effective revocation time (IQR-cleaned, in seconds) as a function of the V2V censoring fraction on a single shared axis across all three schemes. All three use the same last-accepted-message definition (the interval between the revocation decision and the last valid message from the revoked identity accepted by any other vehicle, network-wide), rather than passive being reported on a separate RA-side denial-latency basis as in earlier passes.

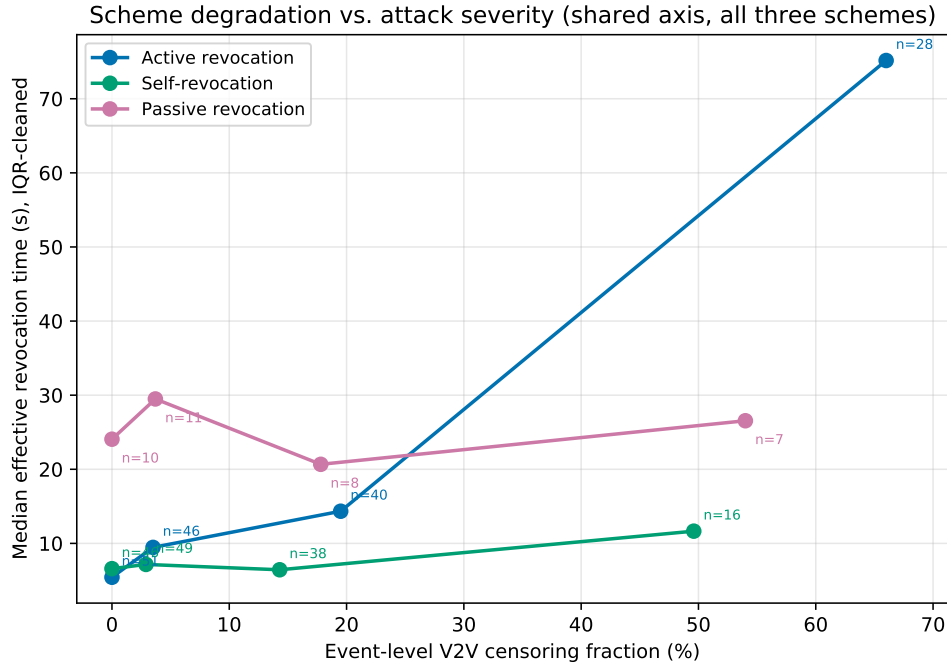


Figure 5.3: Median effective revocation time versus event-level V2V censoring fraction across active, self, and passive revocation schemes under attack A12.

Figure 5.4 tracks the false-positive revocation rate among unrevoked, honest vehicles as

a function of V2V censoring fraction. For active revocation, the rate remains strictly zero as local CRL checks require cryptographically verified RA broadcasts. For self-revocation, the metric records false local revocations induced by clock desynchronization resulting from heartbeat suppression. For passive revocation, the metric measures the percentage of honest vehicles experiencing at least one re-enrollment lockout caused by jamming during pseudonym renewal.

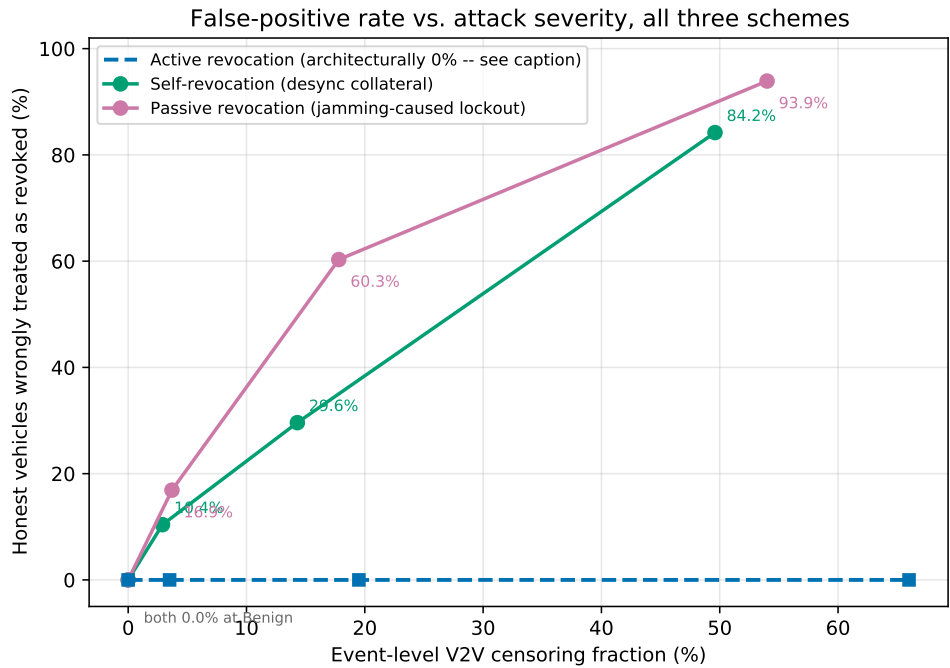


Figure 5.4: False-positive revocation rate versus V2V censoring fraction for honest vehicles under attack A12.

Chapter 6

Analysis & Discussion

6.1 Comparison with state of the art/related works

6.2 Lessons learned

6.3 Limitations of validity

Chapter 7

Future work

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Chapter 8

Conclusions

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